Lab 3: Component Modelling & Architectural Pattern Selection

Self-Service Coffee Kiosk System

SRN: PES1UG24AM396

1. UML Component Diagram

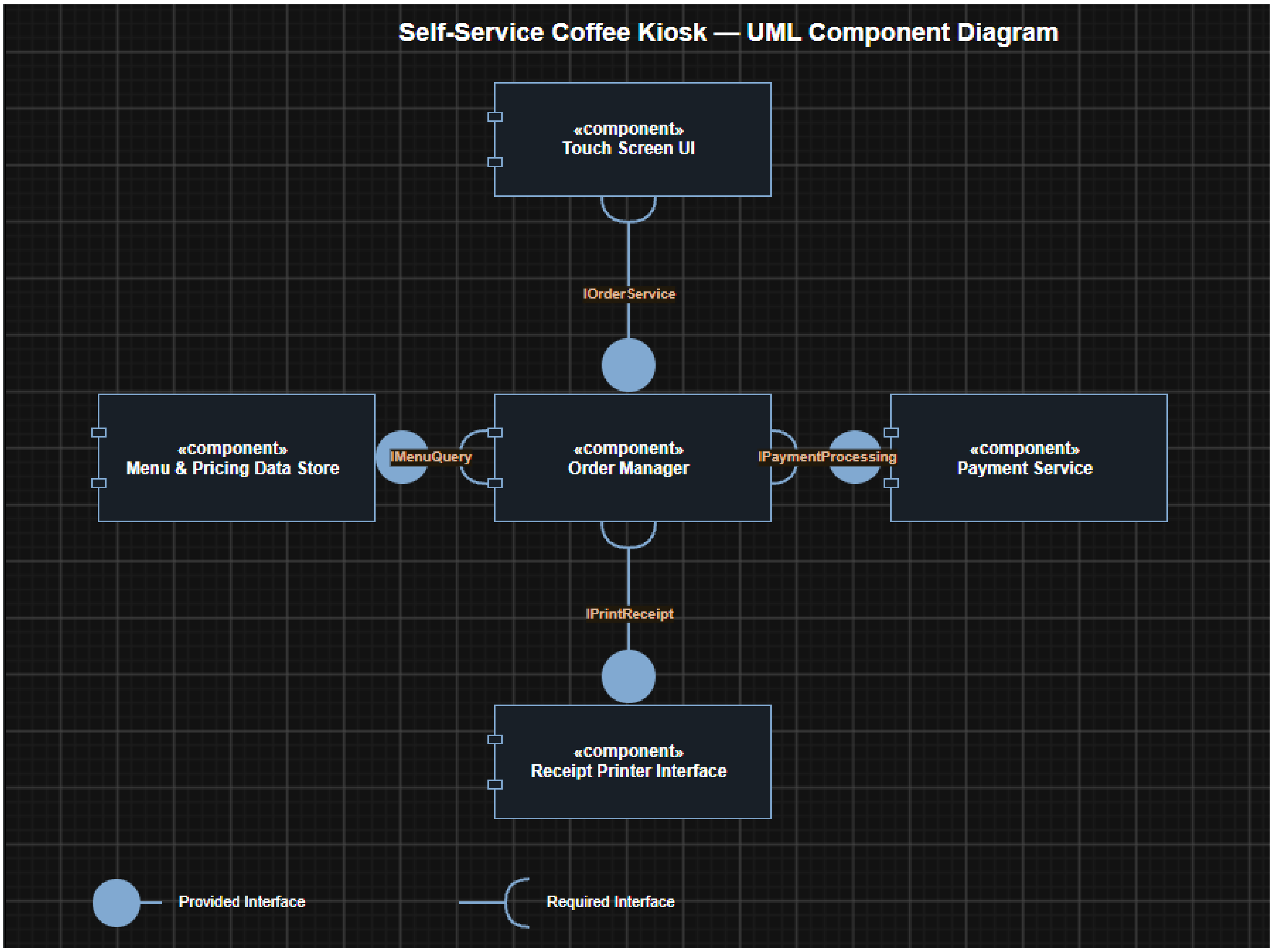

Figure 1 — Component diagram showing 5 components and 4 provided/required interfaces.

2. Written Justification

Architecture Selection

We chose Layered Architecture for the Self-Service Coffee Kiosk System.

Architectural Choice

The system is structured into three layers: a Presentation layer (Touch Screen UI), a Business layer (Order Manager and Payment Service), and a Data/Hardware layer (Menu & Pricing Data Store and Receipt Printer Interface). Each layer only interacts with the layer directly beneath or above it through well-defined interfaces.

Reason 1 — Matches a Single-Device, Fixed-Scope System

The kiosk is a standalone device serving one customer at a time with a fixed, small set of responsibilities (menu selection, payment, printing). Microservices would add network latency and operational complexity (service discovery, independent deployments) that a single-device kiosk with constrained hardware does not need, and a Client-Server split is unnecessary since there is no multi-client access pattern to support.

Reason 2 — Clean Mapping to the Kiosk's Natural Responsibilities

The kiosk's requirements divide naturally into presentation (touch interaction), business logic (order creation, payment processing), and data/hardware access (menu pricing storage, receipt printing). A Layered Architecture keeps these concerns cleanly separated behind component interfaces, which makes the system easier to build, test, and modify (e.g., swapping the receipt printer model) without one change rippling through the whole system.

Security Advantage

By confining all payment-card handling to the Payment Service component within the Business layer, the Presentation layer (Touch Screen UI) never has direct access to raw card data — it only exchanges order totals through the Order Manager's IPaymentProcessing interface. This isolates sensitive payment processing behind a single well-defined boundary, reducing the surface area for tampering or interception at the UI layer.

Performance Benefit

Because the Business layer (Order Manager) can hold menu and pricing data from the IMenuQuery interface in memory for the duration of an order, repeated touchscreen selections during the ordering flow are served from fast in-memory lookups rather than issuing a fresh data-store query for every screen interaction, keeping the UI responsive on the kiosk's constrained hardware.